

Your Title: State the Result, Not the Topic

Your Name*

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Abstract

Replace. One paragraph, at most 150 words: the question, the model in one sentence, the main result *with its condition*, and what the Lean formalization does and does not certify.

1 Introduction

Replace. The question, why it matters, and your answer. State the main result in words in the first page. Say which paper you extend and which assumption you relax (Track A), or which thesis model you formulate (Track B). End with what is new relative to the closest paper. The toy model below only shows the mechanics of this template; none of it is yours to keep.

2 Related literature

Replace. Two or three paragraphs. Position the paper against the closest work, cite the exact result you depart from (proposition and page), and check the editorial status of every preprint you cite.

The task-based view of automation follows [Acemoglu and Restrepo \(2018\)](#); the decision-theoretic view of AI assistance follows [Agrawal et al. \(2025\)](#).

3 Model

Replace. Primitives, timing, information, and the agent’s problem written formally: what is maximised, over which variable, subject to which constraints. Number every assumption; a result may only use numbered assumptions.

A worker chooses effort $e \geq 0$. Output is ae , where a is productivity with AI assistance, and effort costs $\frac{c}{2}e^2$.

Assumption 1. $a > 0$ and $c > 0$.

The worker solves

$$\max_{e \geq 0} ae - \frac{c}{2}e^2. \tag{1}$$

*Universidad del Pacífico. Repository: <https://github.com/your-user/ai-project>. Track A (extension of a course paper) or Track B (thesis model): say which.

4 Results

Replace. Every proposition with *all* its conditions and a complete proof. A proof may be moved to Appendix A, but the statement and the idea of the proof stay here.

Proposition 1. Under Assumption 1, problem (1) has the unique solution $e^* = a/c$, and $\partial e^*/\partial a = 1/c > 0$.

Proof. The objective is twice differentiable with second derivative $-c < 0$, so it is strictly concave and has at most one maximiser. The first-order condition $a - ce = 0$ gives $e = a/c$, which is positive under Assumption 1, so the constraint $e \geq 0$ does not bind and $e^* = a/c$. Differentiating, $\partial e^*/\partial a = 1/c > 0$. $\square$

5 Simulations

Replace. A symbolic or numerical check of every result (SymPy, Monte Carlo, or both), produced by the scripts in `code/`. Say what the simulation can and cannot show: it checks a derivation on a grid, it does not prove it.

Figure 1 compares the closed form of Proposition 1 with a grid search over e . The largest absolute gap is reported by `code/verify.py`, which fails if it exceeds the grid resolution.

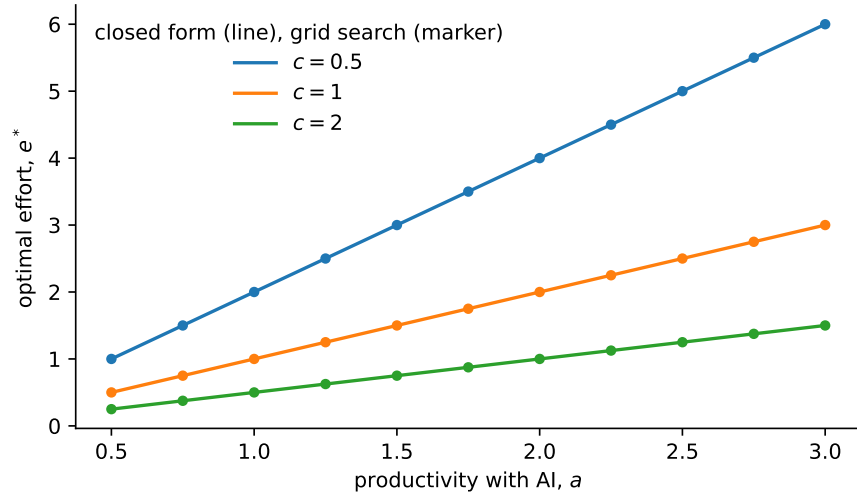


Figure 1: Optimal effort: closed form (lines) and grid search (markers). Source: `code/verify.py`.

6 Discussion and limitations

Replace. What the result depends on, which assumption does the work, and what you could not prove. A stated limitation is worth more than a hidden one.

7 Conclusion

Replace. One paragraph. No new results.

References

Daron Acemoglu and Pascual Restrepo. The race between man and machine: Implications of technology for growth, factor shares, and employment. *American Economic Review*, 108(6): 1488–1542, 2018. doi: 10.1257/aer.20160696.

Ajay Agrawal, Joshua Gans, and Avi Goldfarb. The economics of bicycles for the mind. Working Paper 34034, National Bureau of Economic Research, 2025.

A Omitted proofs

Replace. Proofs and derivations moved out of the body. The handwritten, step-by-step version of every derivation lives in `hand/`.

B Lean formalization

Replace. One row per numbered result of the paper. “Status” is one of: *proved* (no `sorry`, no added hypothesis), *proved under an added hypothesis* (state it), or *open* (state the exact blocker). Report the paper commit that was formalized and the output of `paper_contribution.py check <Folder> -fast`.

Paper	Lean declaration	File in <code>lean/</code>	Status
Proposition 1	<code><Folder>.prop_effort</code>	<code>MainTheorems.lean</code>	<i>example row</i>

C AI collaboration log

Replace. For each relevant contribution of a language model: what it claimed, your verdict (correct, incorrect, or not verifiable) with the reason, and how you checked it. The raw prompts and answers are in `prompts.md`.

Claim by the model	Verdict	How it was checked
<i>example:</i> “ $e^* = a/c$ for any a ”	Incorrect as stated	Fails for $a \leq 0$: the corner $e^* = 0$ binds. Hand derivation, p. 1.